1. Let $A = \{0, 2, 4, 6, 8, 9, 10\}$ and $B = \{1, 3, 5, 7, 9\}$. Which one of the following is true?
   A. $B - A = B$
   B. $A - B = A$
   C. $B - A = \{1, 3, 5, 7\}$
   D. $A - B = \{0, 2, 4, 6, 8, 10\}$
2. Consider the polynomial function $f(x) = ax^4 - b$, where a and b are real numbers. Which one of the following is true about f?
   A. The graph of f opens upward for any $b > 0$.
   B. The graph of f opens upward for any $a > 0$.
   C. The graph of f opens downward for any $b \leq 0$.
   D. The graph of f opens downward for any $a \leq 0$.
3. The solution set of the inequality $3x^2 + 35x - 12 < 0$ is
   A. $(-12, \frac{1}{3})$
   B. $(-\infty, \frac{1}{3})$
   C. $(-12, 1)$
   D. $(-\infty, -12)$
4. A closed right circular cone has a base radius $2\ \text{cm}$ and slant height $3\sqrt{2}\ \text{cm}$. What is the total surface area of the cone?
   A. $2\pi(1 + 3\sqrt{2})\ \text{cm}^2$
   B. $2\pi(2 + 3\sqrt{2})\ \text{cm}^2$
   C. $6\pi\sqrt{14}\ \text{cm}^2$
   D. $2\pi\sqrt{14}\ \text{cm}^2$
5. The lower and upper bases of the frustum of a regular pyramid are equilateral triangles of side $3\ \text{cm}$ long and $2\ \text{cm}$ long respectively. If its slant height is $4\ \text{cm}$, then what is the lateral surface area of the frustum?
   A. $30\ \text{cm}^2$
   B. $48\ \text{cm}^2$
   C. $72\ \text{cm}^2$
   D. $96\ \text{cm}^2$
6. Which one of the following is true about the signum function, $f(x) = \operatorname{sgn} x$?
   A. Its domain is the set of real numbers.
   B. Its range is the set of real numbers.
   C. Its domain is $\{-1, 0, 1\}$
   D. Its range is $[0, \infty)$
7. Which one of the following is the equation of a circle with center $(2, -1)$ and radius $3\ \text{units}$?
   A. $x^2 + y^2 + 2x + 2y - 4 = 0$
   B. $x^2 + y^2 - 4x + 2y - 4 = 0$

C. $x^2 + y^2 + 4x - 2y - 5 = 0$

D. $x^2 + y^2 - 4x + 2y - 5 = 0$

8. Consider the ellipse given by the equation $4(x+1)^2 + (y-1)^2 = 16$. Which one of the following is the standard form of equation of the ellipse?

A. $\frac{(x+1)^2}{4^2} + \frac{(y-1)^2}{1^2} = 1$

B. $\frac{(x+1)^2}{4^2} + \frac{(y-1)^2}{4^2} = 1$

C. $\frac{(x+1)^2}{2^2} + \frac{(y-1)^2}{4^2} = 1$

D. $\frac{(x+1)^2}{2^2} + \frac{(y-1)^2}{1^2} = 1$

9. Which of the following is true? A hyperbola is a locus of points in the plane so that

A. the sum of the distances from two fixed points is constant.

B. the difference between the distances from two fixed points is constant.

C. the points have the same distance from a given point and a given line.

D. the points have the same distance from any given two points.

10. Which of the following is the converse of, "If wages increase, then inflation decreases."?

A. If inflation decreases, then wages increase.

B. If inflation decreases, then wages decrease.

C. If inflation increases, then wages decrease.

D. If inflation increases, then wages increase.

11. Let $A = \begin{pmatrix} 0 & -1 \\ 5 & 1 \end{pmatrix}$. Which one of the following is equal to $A + A^T$?

A. $\begin{pmatrix} 0 & -4 \\ -4 & 2 \end{pmatrix}$

B. $\begin{pmatrix} 0 & 4 \\ 4 & 2 \end{pmatrix}$

C. $\begin{pmatrix} 0 & 5 \\ 5 & 1 \end{pmatrix}$

D. $\begin{pmatrix} 0 & -5 \\ -5 & -1 \end{pmatrix}$

12. Consider a line with equation $y = mx + b$. Which of the following is true?

A. The line intersects the y axis, if $m \neq 0$ and $b > 0$.

B. The line intersects the x axis, if $m = 0$ and $b > 0$.

C. The line has no slope.

D. The line passes through the origin.

13. Which one of the following statements is NOT true about the graph of the given linear inequality?

A. The graph of the linear inequality $y > 2x - 1$ contains all points of the line $y = 2x$

B. The graph of the linear inequality $y \leq 2x + 1$ is below the line $y = x$ on the interval $(-\infty, -1)$

C. The graph of the linear inequality $y > 2x + 1$ contains no points of the line $y = 2x - 3$

D. The graph of the linear inequality $y \leq x - 1$ is above the line $y = 2x$ on the interval $(-\infty, -1)$

14. Voter turnout in the local school board elections declined 40% from last year. This year only 3600 eligible voters went to the polls. How many people voted in last year's school board elections?
    A. 6,000
    B. 5,500
    C. 5,000
    D. 4,500
15. You happened to notice that the 8% sales tax that your father paid on the new Television Set came to Birr 3,800. How much did the Television Set cost him?
    A. Birr 47,500
    B. Birr 45,800
    C. Birr 35,400
    D. Birr 30,600
16. What is the simplified form of $\frac{\sqrt[3]{32}}{16^{\frac{2}{3}}}$?
    A. 2
    B. $\frac{1}{2}$
    C. 8
    D. $\frac{1}{8}$
17. Which one of the following numbers is equal to $\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$

A. $6 + 2\sqrt{2}$

B. $5 + 2\sqrt{6}$

C. $1 + 2\sqrt{3}$

D. $3 - 2\sqrt{3}$

18. What is the solution set of the equation $6 - 4x - 2x^2 = 0$?
    A. $\{-6, 2\}$
    B. $\{-2, 6\}$
    C. $\{-3, 1\}$
    D. $\{-1, 3\}$
19. Let $f(x) = x^2 - x$ and $g(x) = x + 1$. Which one of the following statements is true?
    A. $(f + g)(-2) = 3$
    B. $(f - g)(-2) = 5$
    C. $(f \cdot g)(-2) = -5$
    D. $(\frac{f}{g})(-2) = -6$

20. What is the area of an equilateral triangle inscribed in a circle of radius 4 units?
    A. $6\sqrt{3}$ sq. units
    B. $12\sqrt{3}$ sq. units
    C. 12 sq. units
    D. 36 sq. units
21. Consider ΔABC with right angle at C and $m(\angle ABC) = 55°$. If the length of $\overline{AC}$ is 10 units, then what is the area of the triangle?
    (Given $\tan 35° = 0.7$ and $\tan 55° = 1.43$)
    A. 35 sq. units
    B. 70 sq. units
    C. $5\sqrt{149}$ sq. units
    D. $10\sqrt{149}$ sq. units
22. The base of a right prism is a rectangle with sides 3 cm and 4 cm and the altitude of the prism is 5 cm. What is the total surface area of the prism?
    A. $70\ \text{cm}^2$
    B. $82\ \text{cm}^2$
    C. $94\ \text{cm}^2$
    D. $102\ \text{cm}^2$
23. Let $\vec{u}$ be a vector of length $3\sqrt{3}$ cm directed to the East and $\vec{v}$ be a vector of length 3 cm directed to the north. Then $\vec{u} + \vec{v}$ is equals to a vector of length:
    A. 3 cm in the direction East 60° North.
    B. 3 cm in the direction East 30° North.
    C. 6 cm in the direction East 60° North.
    D. 6 cm in the direction East 30° North.
24. Below is a histogram showing the time taken to complete a test.
    Based on the above histogram, which of the following is true?
    A. 30% of the students completed the test with less than 40 minutes.
    B. 25% of the students completed the test with more than 50 minutes.
    C. 50% of the students completed the test with less than 50 minutes.
    D. 40% students completed the test with more than 40 minutes.
25. Consider the graphs of the functions $y = a^x$ and $y = \log_a x$ for $a > 0$ and $a \neq 1$. Which of the following is true about the functions?
    A. The graph of $y = a^x$ is a reflection of the graph of $y = \log_a x$ about the origin.
    B. The graph of $y = a^x$ is a reflection of the graph of $y = \log_a x$ along $y = -x$.
    C. For $a > 1$, the graphs of both $y = a^x$ and $y = \log_a x$ are increasing on $(0, \infty)$.
    D. For $0 < a < 1$ the y-axis is an asymptote to the graphs of $y = a^x$ and $y = \log_a x$.
26. What is the solution set of the equation $3^{x-x^2} = \frac{1}{9}$?
    A. $\{0, 1\}$

B. $\{-1, 0\}$
C. $\{1, -2\}$
D. $\{-1, 2\}$

27. Which of the following is true about the trigonometric values of 855°?
A. $\cos 855^\circ = -\frac{\sqrt{2}}{2}$
B. $\sin 855^\circ = \frac{\sqrt{3}}{2}$
C. $\tan 855^\circ = 1$
D. $\cot 855^\circ = -\sqrt{2}$

28. Which one of the following is true about the function $y = \sin x$?
A. Its domain is $[-1, 1]$
B. Its period is π
C. Its range is $[-1, 1]$
D. Its domain is $(0, \infty)$

29. Let $f(x) = \frac{3x+1}{x+2}$. Which one of the following is the inverse of f?
A. $f^{-1}(x) = \frac{2x-1}{x-3}$
B. $f^{-1}(x) = \frac{-2x+1}{x-3}$
C. $f^{-1}(x) = \frac{x-2}{-3x-1}$
D. $f^{-1}(x) = \frac{x+2}{3x+1}$

30. Suppose that the lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{BC}$ intersect at B outside the circle as shown in the figure below. If $m(\angle ABC) = 30^\circ$ and $m(\angle BCD) = 45^\circ$, then what is $m(\widehat{AC})$?
A. 90°
B. 105°
C. 120°
D. 150°

31. Consider the rational function $f(x) = \frac{x^3-1}{x^2+1}$. Which one of the following is true about the graph of f?
A. Its y intercept is -1
B. It has no x intercept
C. It is symmetrical about the y axis.
D. It is symmetrical about the origin.

32. If two cards are drawn one after the other from a well shuffled pack of cards, then which one of the following is an independent event?
A. Drawing a king in the first draw and a queen in the second draw without replacing the first.
B. Drawing a king in the first draw and a queen in the second draw by replacing the first.
C. Drawing kings in both draws without replacing the first.
D. Draw queens in both draws without replacing the first.

33. Consider the grouped frequency table for students' weight in a certain school.

Weight (kg)	Frequency
31-35	3
36-40	5
41-45	6
46-50	7
51-55	4

What is the mean weight of the students?

A. 41.5 kg

B. 43 kg

C. 43.8 kg

D. 45.5 kg

34. A bag contains 2 red, 3 white and 1 black balls. We draw two balls one after the other without replacement. What is the probability that the first ball is red and the second ball is black?
    A. $\frac{1}{10}$
    B. $\frac{1}{12}$
    C. $\frac{1}{15}$
    D. $\frac{1}{18}$
35. Let $A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$ and $\det(A) = 3$. Which one of the following is NOT true?
    A. $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \\ a_2 & b_2 & c_2 \end{vmatrix} = -3$
    B. $\begin{vmatrix} 2a_1 & 2b_1 & 2c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 6$
    C. $\begin{vmatrix} -a_1 & b_1 & -c_1 \\ -a_2 & b_2 & -c_2 \\ -a_3 & b_3 & -c_3 \end{vmatrix} = 3$
    D. $\begin{vmatrix} a_1 & b_1 & b_1 \\ a_2 & b_2 & b_2 \\ a_3 & b_3 & b_3 \end{vmatrix} = -3$
36. Let $z_1 = 2 - 3i$ and $z_2 = 5 + 7i$. Which one of the following is equal to $\frac{z_1}{z_2}$?
    A. $-\frac{14}{46} + \frac{20}{46}i$
    B. $-\frac{11}{46} + \frac{10}{46}i$

C. $-\frac{11}{74} - \frac{29}{74}i$

D. $\frac{31}{74} - \frac{29}{74}i$

37. Which of the following is NOT true?
    A. There is no single rule for determining the size of a sample.
    B. Random sampling requires time to plan and implement.
    C. Random sampling is free from personal bias of the investigator.
    D. Any person can implement random sampling from a given population.
38. Consider the following data: 6, 6, 7, 7, 7, 8, 8. Then which of the following is true?
    A. The mean deviation from the mode is $\frac{4}{7}$.
    B. The mean deviation from the median is $\frac{7}{4}$.
    C. The mean deviation from the mean is $\frac{7}{4}$.
    D. The mean deviation from the mean is $\frac{3}{2}$.
39. Consider two firms with the given mean income and standard deviations.

Firm	Mean income (in Birr)	Standard deviation (in Birr)
Firm I	1,500,000	300
Firm II	1,500,000	350

Which of the following is true?

A. The data set in Firm I have greater variability than Firm II.

B. The data set in Firm I have less variability than Firm II.

C. The two data sets have the same consistency.

D. We cannot compare the two data sets.

40. One special perfume costs a cosmetic shop Birr 2,100. If the shop desires a 25% markup rate based on the selling price, then which one of the following should be set as a selling price for this perfume?
    A. Birr 2,800
    B. Birr 2,600
    C. Birr 2,400
    D. Birr 2,300
41. A couple own a house that values Birr 2,000,000. The house added Birr 10,000 in the first year, Birr 10,400 in the second year, Birr 10,800 in the third year and so on. What is the value of the house at the end of 6 years?
    A. Birr 2,012,000
    B. Birr 2,054,000

C. Birr 2,060,000

D. Birr 2,064,000

42. Let f be a function defined on an open interval about a real number a, except possibly at a itself. Which one of the following is necessary for $\lim_{x \to a} f(x)$ to exist?
    A. $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x)$
    B. $\lim_{x \to a^+} f(x) = f(a)$
    C. $\lim_{x \to a^+} f(x)$ exists
    D. $\lim_{x \to a^-} f(x)$ exists
43. Let f be a function that is continuous at x_0. Which of the following is true about f?
    A. x_0 does not belong to the domain of f
    B. $\lim_{x \to x_0} f(x)$ does not exist
    C. $\lim_{x \to x_0^+} f(x) \neq \lim_{x \to x_0^-} f(x)$
    D. $\lim_{x \to x_0} f(x) = f(x_0)$
44. Let $f(x) = 2\sqrt{x+1}$. Which one of the following indicates the interval and its derivative of f respectively?
    A. $[0, \infty), (x+1)^{\frac{3}{2}}$
    B. $[-1, \infty), \frac{1}{\sqrt{x+1}}$
    C. $(-1, \infty), \frac{1}{\sqrt{x+1}}$
    D. $(-\infty, -1), (x+1)^{\frac{1}{2}}$
45. Let $f(x) = x^2 + \sqrt{x}$ and $g(x) = e^{2x}$. Which of the following is true?
    A. $(f+g)'(x) = 2x + \frac{1}{2\sqrt{x}} + 2e^{2x}$
    B. $(f+g)'(x) = 2x + \frac{1}{2}\sqrt{x} + 2e^{2x}$
    C. $(f+g)'(x) = 2x - \frac{1}{2}\sqrt{x} - 2e^{2x}$
    D. $(f-g)'(x) = 2x - \frac{1}{2\sqrt{x}} - 2e^{2x}$
46. What is the equation of the tangent line to the graph of the function $f(x) = e^x$ at $(1, e)$?
    A. $y = x + \frac{1}{e}$
    B. $y = \frac{1}{e}x + \frac{1}{e}$
    C. $y = ex + e$
    D. $y = ex$
47. Water is flowing into a cylindrical water tank of radius 2m and height 10m at a rate of $5\ \mathrm{m}^3/\mathrm{min}$. What is the rate of the change of the height when the water level is 6m?
    A. $\frac{5}{4\pi}\ \mathrm{m/min}$
    B. $\frac{5\pi}{4}\ \mathrm{m/min}$
    C. $2\ \mathrm{m/min}$
    D. $2\pi\ \mathrm{m/min}$
48. Which one of the following is equal to $\int (e^{2x} + \cos x)\, dx$?
    A. $\frac{1}{2}e^{2x} + \sin x + c$

B. $\frac{1}{2}e^{2x} - \sin x + c$

C. $e^{2x} + \cos x + c$

D. $e^{2x} + \sin x + c$

49. Let f be continuous on $[a, b]$ and $F'(x) = f(x)$ for all $x \in [a, b]$. Which of the following is true?

A. $\int_a^b F(x)\,dx = f(b) - f(a)$

B. $\int_a^b F(x)\,dx = f(a) - f(b)$

C. $\int_a^b f(x)\,dx = F(a) - F(b)$

D. $\int_a^b f(x)\,dx = F(b) - F(a)$

50. Which one of the following is equal to $\int_1^3 2x\sqrt{x^2+1}\,dx$?

A. $\frac{4(10\sqrt{10}-2\sqrt{2})}{3}$

B. $\frac{2(5\sqrt{10}-\sqrt{2})}{3}$

C. $\frac{2(5\sqrt{10}+2\sqrt{2})}{3}$

D. $\frac{4(5\sqrt{10}-\sqrt{2})}{3}$

51. A line makes an angle of 145° with the positive x-axis. What is the slope of the line? (Given $\tan 35^\circ = 0.7$)

A. $\frac{10}{7}$

B. $-\frac{10}{7}$

C. 0.7

D. -0.7

52. What is the solution set of $\frac{2}{x-1} + \frac{4}{x} = \frac{6}{x-2}$?

A. $\left\{\frac{4}{11}\right\}$

B. $\left\{\frac{11}{4}\right\}$

C. $\left\{\frac{4}{5}\right\}$

D. $\left\{\frac{5}{4}\right\}$

53. Let $l_1 : y = \frac{1}{3}x + 1$ and $l_2 : y = 2x - 1$ be two lines. What is the angle between l_1 and l_2 measured from l_1 to l_2 counterclockwise?

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. $\frac{3\pi}{4}$

D. $\frac{3\pi}{2}$

54. Let p: 2 is a prime number and q: 3 is an odd number. Then the contrapositive and its truth value of the statement $p \Rightarrow q$ are respectively

A. $\neg q \Rightarrow \neg p$ and True

B. $\neg q \Rightarrow \neg p$ and False

C. $\neg p \Rightarrow q$ and True

D. $q \Rightarrow \neg p$ and False

55. The following table shows the test scores of 20 students in a class.

Test Scores	Frequency
0-2	3
3-5	5
6-8	5
9-11	3
12-14	4

What is the variance of the given data?

A. 16.2

B. 17.2

C. 18.45

D. 19.45

56. What is the solution of the system of linear equations $\begin{cases} 2x + 7y = -5 \\ 5x + 11y = 2 \end{cases}$?

A. $x = -\frac{5}{3}$ and $y = \frac{25}{21}$

B. $x = \frac{5}{3}$ and $y = -\frac{25}{21}$

C. $x = \frac{69}{13}$ and $y = -\frac{29}{13}$

D. $x = -\frac{69}{13}$ and $y = \frac{29}{13}$

57. What is the simplified form of $\frac{(2+i)(3-4i)}{(2-2i)(1+i)}$?

A. $\frac{7}{8} - \frac{3}{8}i$

B. $\frac{5}{2} - \frac{5}{2}i$

C. $\frac{5}{2} + \frac{5}{4}i$

D. $\frac{5}{2} - \frac{5}{4}i$

58. Consider the sequence $\{a_n\}_{n=1}^{\infty}$, where $a_n = 1 + \frac{(-1)^n}{n}$ for $n \geq 1$. Which one of the following is true?

A. $a_{10} = \frac{11}{10}$

B. $a_{11} = \frac{12}{11}$

C. $a_n \leq a_{n+1}$ for all n

D. $a_n \geq a_{n+1}$ for all n

59. Consider the sequence $\{1 + \frac{1}{n}\}_{n=1}^{\infty}$. Which one of the following is true about the sequence?

A. 2 is its lower bound

B. $\frac{3}{2}$ is its lower bound

C. 4 is its upper bound.

D. $\frac{6}{5}$ is its upper bound

60. Which one of the following is equal to $\lim_{x\to 0} \frac{x}{1-\sqrt{x+1}}$?

    A. 0

    B. -2

    C. -1

    D. $-\frac{1}{2}$

61. Let $f(x) = e^{-2x}$. Which one of the following is true about f?

    A. $f^{(n)}(0) = (-\frac{1}{2})^n$

    B. $f^{(n)}(0) = (-\frac{1}{2e})^n$

    C. $f^{(n)}(x) = (-2)^n e^{-2x}$

    D. $f^{(n)}(x) = 2^n e^{-2x}$

62. Consider the function $f(x) = x^2 - 3x$ on $[-1, 2]$. What is the value x that satisfies the Mean Value Theorem?

    A. 0

    B. 1

    C. $\frac{1}{2}$

    D. $\frac{3}{2}$

63. A carpenter wants to design a closed box that has a square base and total surface area of $54\ \text{cm}^2$. What is the maximum volume that can be constructed with the given total surface area?

    A. $12\ \text{cm}^3$

    B. $13.5\ \text{cm}^3$

    C. $18\ \text{cm}^3$

    D. $27\ \text{cm}^3$

64. What is the area of the region bounded by the line $y = x$ and the parabola $y = x^2 - 2x$ on the interval $[0, 3]$?

    A. 9 square units

    B. $\frac{9}{2}$ square units

    C. 3 square units

    D. 2 square units

65. A painter buys 5 liters of paint for Birr 36 and 6 liters of the same brand paint for Birr 43. Which of the following is true?

    A. Buying 5 liters is more economical than buying 6 liters.

    B. Buying 6 liters is more economical than buying 5 liters.

    C. Both have the same economical purchase.

    D. We cannot compare.